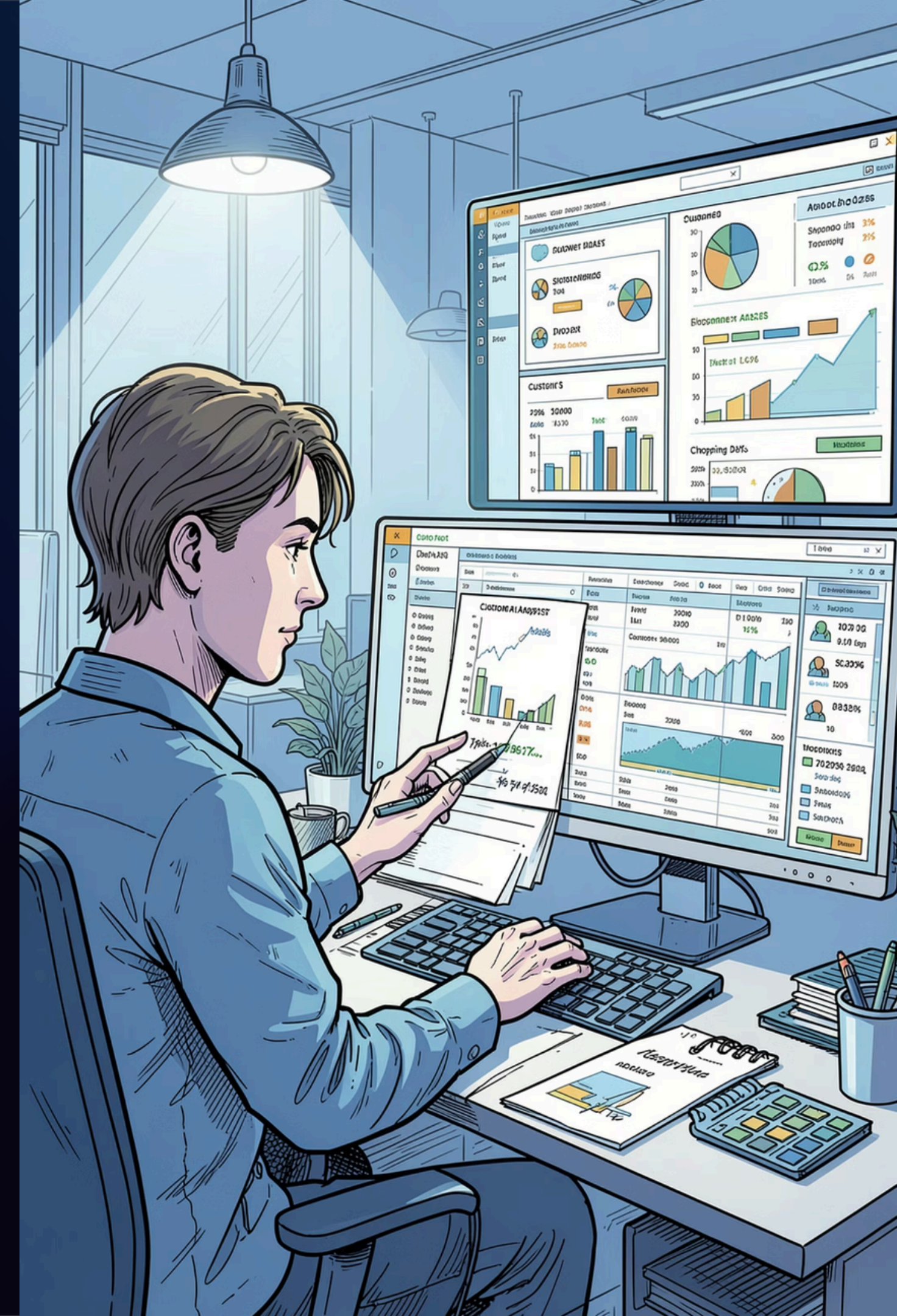


CUSTOMER BEHAVIOR DASHBOARD



Customer Shopping Behavior From Raw Data to Business Insights

An end-to-end analysis of customer shopping data – exploring, cleaning, enriching, and querying to uncover patterns in spending, loyalty, and product performance.



Cleaning the Dataset for Reliable Analysis

Three targeted steps improved data quality before any analysis began.

Missing Review Ratings

Gaps in Review Rating were filled with the **median rating per category**, preserving category-level patterns.

Standardized Column Names

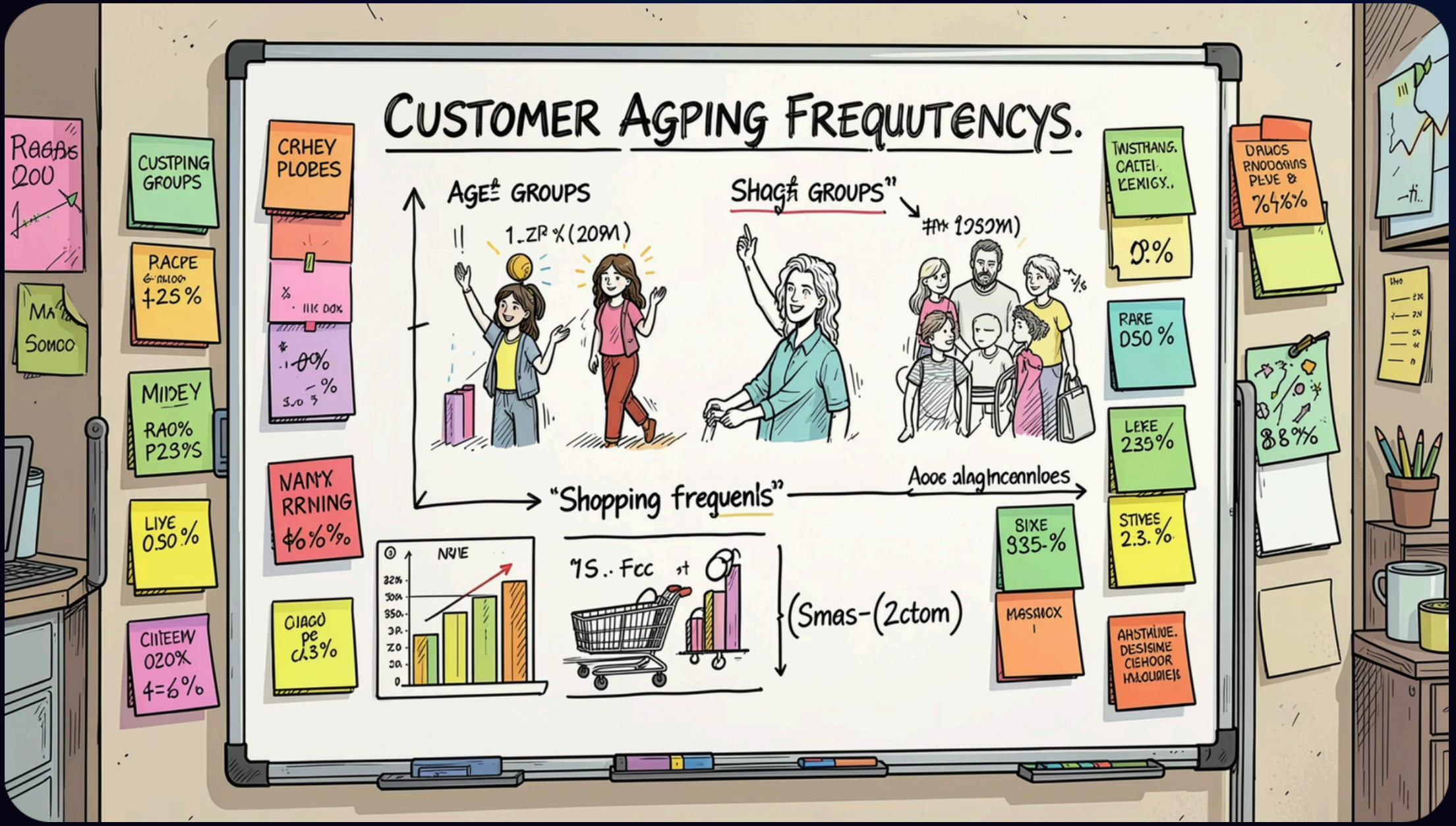
Fields were converted to **snake_case** for consistency. `purchase_amount_(usd)` was simplified to `purchase_amount`.

Redundancy Removal

`discount_applied` and `promo_code_used` were duplicates – the latter was **removed** to streamline the dataset.

Two New Features, Far Richer Insights

Two engineered features transformed raw fields into analytically powerful dimensions.



Age Group

Customers segmented into **Young Adult**, **Adult**, **Middle-aged**, and **Senior** quartiles — enabling behavior comparisons across life stages.

Purchase Frequency (Days)

Text labels converted to numeric intervals: **Weekly** → **7 days**, **Monthly** → **30**, **Annually** → **365** — enabling quantitative frequency analysis.

Answering Real Business Questions

The cleaned dataset was loaded into a customer table and queried using aggregates, CTEs, and window functions to surface patterns across ten business questions.

REVENUE



- Revenue by Gender & Age Group
- Shipping Cost vs Spending

LOYALTY



- New, Returning & Loyal Segmentation
- Repeat Buyers vs Subs
- Subscription Value

PRODUCTS



- Top-Rated Items
- Promo-Driven Products
- Top Category Performers

PROMOTIONS



- High-Value Discount Users

What the Queries Revealed

- **Revenue:** Compared by gender, age group, and shipping choice
- **Loyalty:** Segmented customers as New, Returning, or Loyal; assessed subscription value
- **Products:** Ranked top-rated and promotion-driven items per category
- **Promotions:** Identified high-value buyers who still use discounts

DASHBOARDING

Visualizing Insights with Power BI

Leveraging DAX, the cleaned data was transformed into interactive dashboards, making complex patterns accessible for business decision-making.



DAX Measures & Columns

Calculated total customers, average review ratings, and dynamic revenue/sales breakdowns.



Key Metric Visualizations

Dashboard spotlights revenue by category/age, total customers, and overall sales performance.



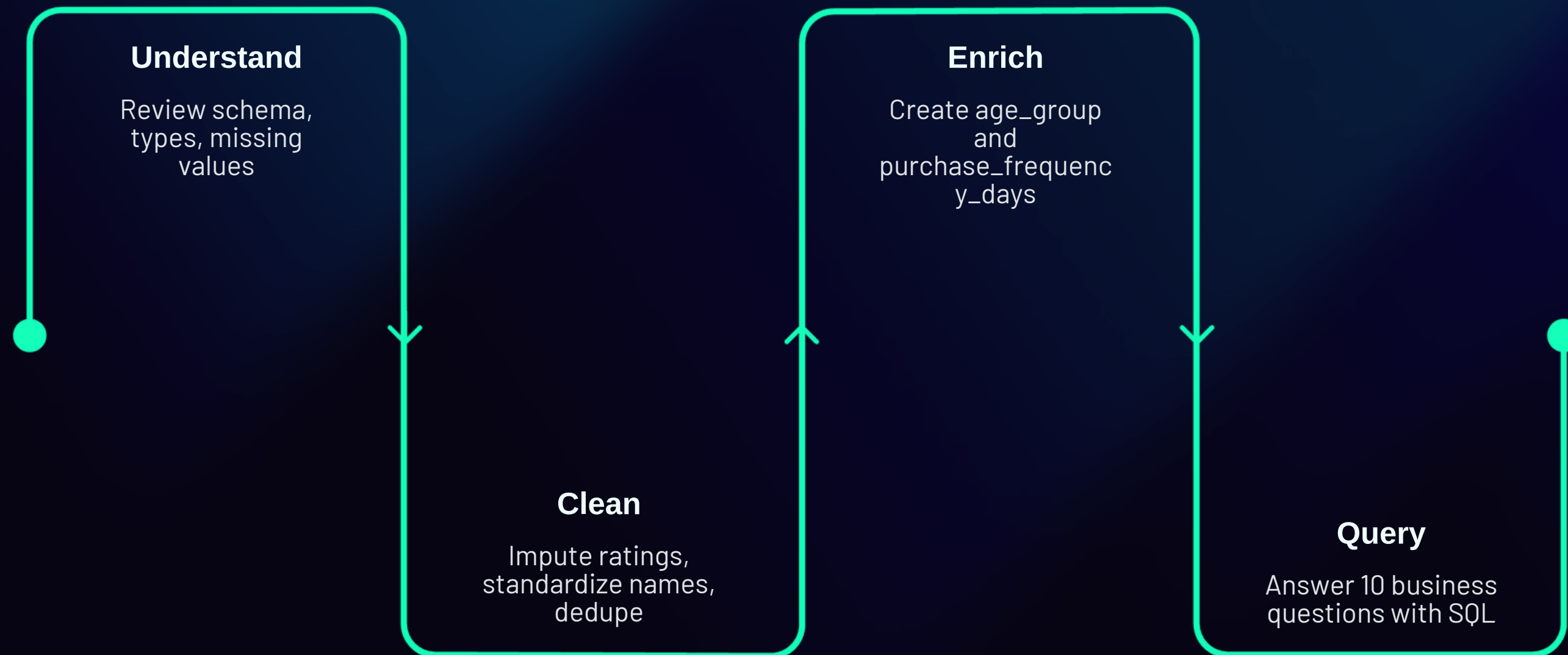
Interactive Filtering

Exploration by subscription status, gender, and product category for deeper analysis.



SUMMARY

A Coherent Pipeline from Raw Records to Ready Insights



Each stage builds on the last — from initial data review through cleaning, enrichment, and finally SQL queries — producing a clear, business-ready view of customer behavior.

Cleaner Data

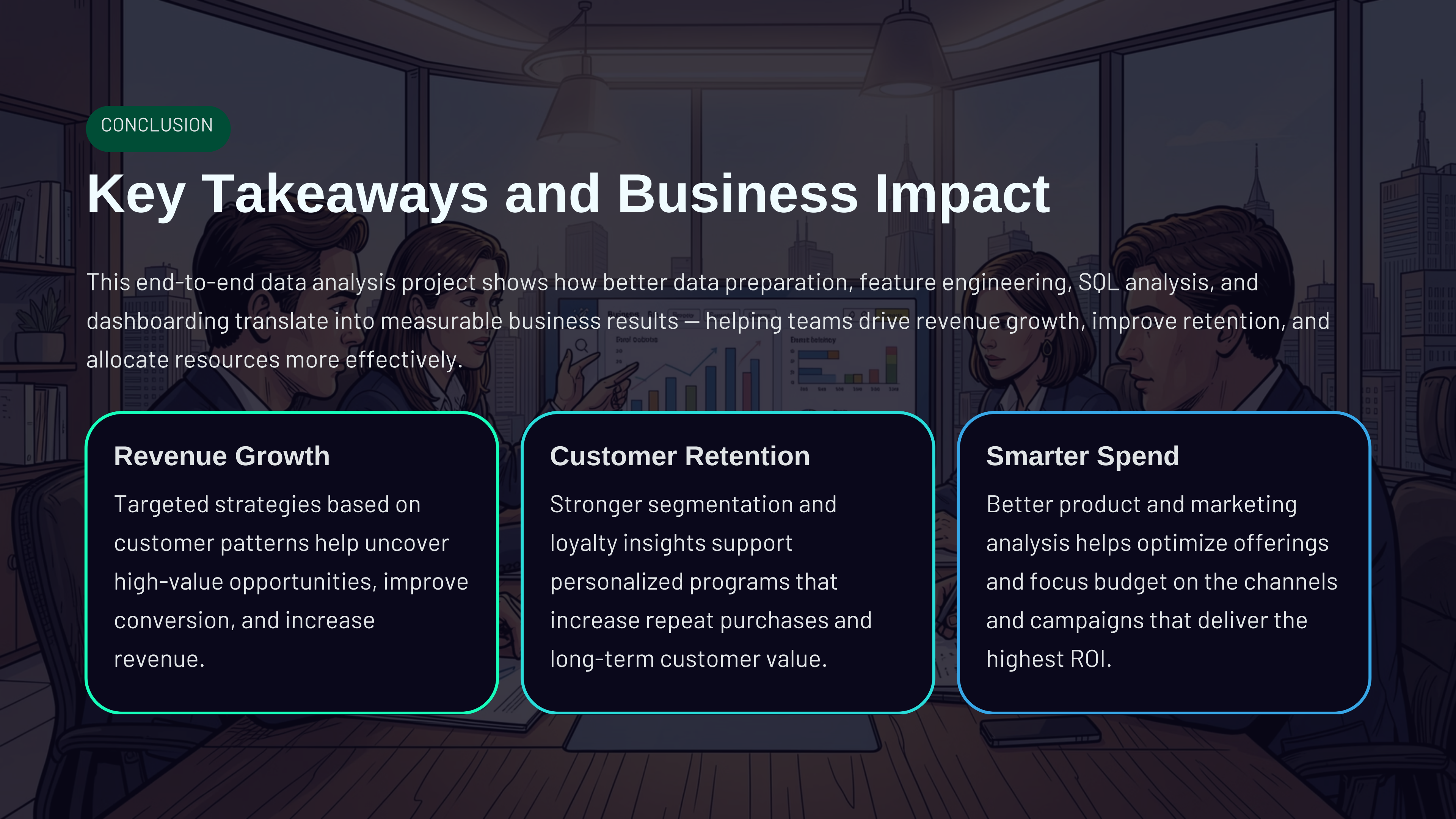
Missing ratings handled, redundant columns removed

Richer Features

Age groups and frequency days unlock new comparisons

Actionable Insights

Revenue, loyalty, and product patterns surfaced through SQL



CONCLUSION

Key Takeaways and Business Impact

This end-to-end data analysis project shows how better data preparation, feature engineering, SQL analysis, and dashboarding translate into measurable business results — helping teams drive revenue growth, improve retention, and allocate resources more effectively.

Revenue Growth

Targeted strategies based on customer patterns help uncover high-value opportunities, improve conversion, and increase revenue.

Customer Retention

Stronger segmentation and loyalty insights support personalized programs that increase repeat purchases and long-term customer value.

Smarter Spend

Better product and marketing analysis helps optimize offerings and focus budget on the channels and campaigns that deliver the highest ROI.